CUSTOMER BEHAVIOR ANALYSIS

E-Commerce Strategic Report

Segmentation · Retention · Growth Insights

350	226	124	0
Total Customers	Low Risk (Active)	Medium Risk (At-Risk)	High Risk

1. Executive Summary

This report is an in-depth review of customer behavior in an e-commerce platform whose data was analyzed on a sample of 350 customers. The main goals were to investigate the data, profile customers, find out the major sources of satisfaction, and create focused retention plans.

Key findings at a glance:

- Gold members and customers in major metropolitan areas such as New York are the high-value customer segments — they spend nearly 3x more than Bronze members and are the most satisfied.
- There is a strong relationship between customer satisfaction and expenditure patterns — happier customers consistently spend more and buy more items.
- 35.4% of customers (124) are classified as Medium Risk of churn — the single greatest growth opportunity if re-engaged promptly.
- No high-risk customers were found in the current dataset, which is a positive signal — the business still has time to act before customers fully disengage.
- Houston and Chicago significantly lag behind top markets in both spend and satisfaction, pointing to clear gaps in regional marketing and customer experience.
- Young professionals aged 28–30 are the highest spenders, making them a prime target for loyalty investment and premium campaigns.

2. Data Exploration & Preparation

The dataset ecommerce_dataset.csv was loaded into a Pandas DataFrame successfully. It includes 350 records and 11 columns. Below are the key findings from the data quality and preprocessing stage.

Check	Finding / Action Taken
Dataset Shape	350 records × 11 columns with a mix of integers, floats, and objects.
Missing Values	2 missing values found in Satisfaction Level — filled with the mode (most frequent value) to ensure data integrity.

Duplicates	No duplicate rows detected.
Outlier Detection	Boxplots for Total Spend and Items Purchased show no extreme outliers indicating consistent data. Average Rating is slightly left-skewed but within expected bounds.
Feature Engineering	Categorical variables (Gender, City, Membership Type, Satisfaction Level) successfully converted to numeric format using LabelEncoder — essential for future quantitative modeling.

Outlier Analysis — Boxplots

Boxplots were generated for three key numerical features to identify distribution characteristics and potential outliers:

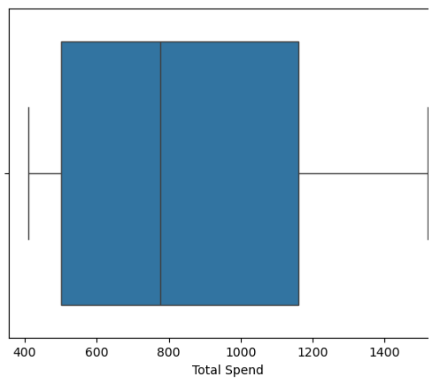

Fig 1. Total Spend

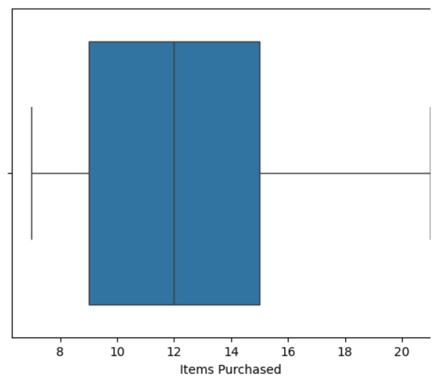

Fig 2. Items Purchased

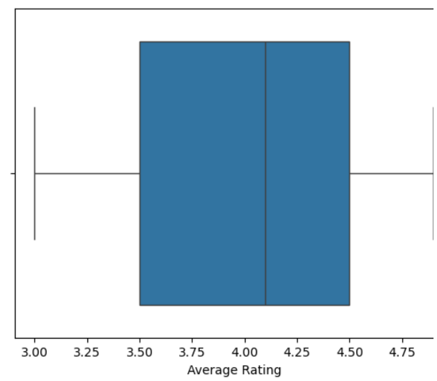

Fig 3. Avg Rating

3. Customer Segmentation

Customers were identified in groups based on demographics (gender, age), membership status, and geographic location (city). The following analysis reveals clear patterns in spending patterns and satisfaction levels.

3.1 Gender Segmentation

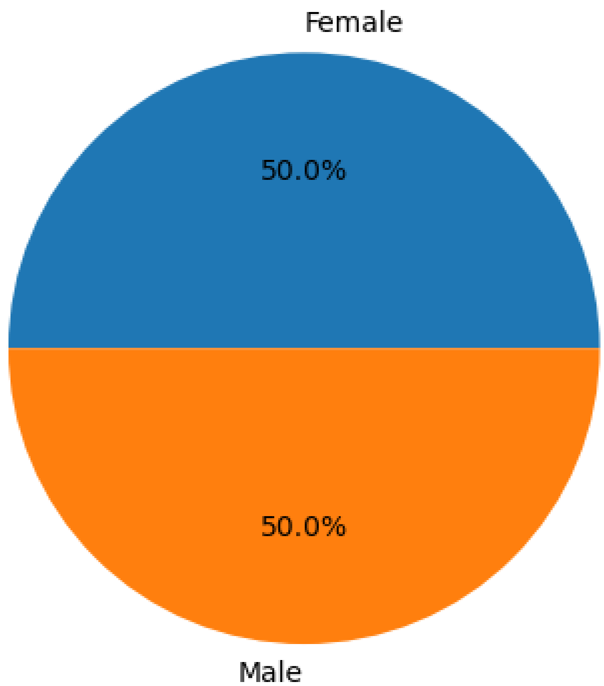

Fig 4. Customer Distribution by Gender

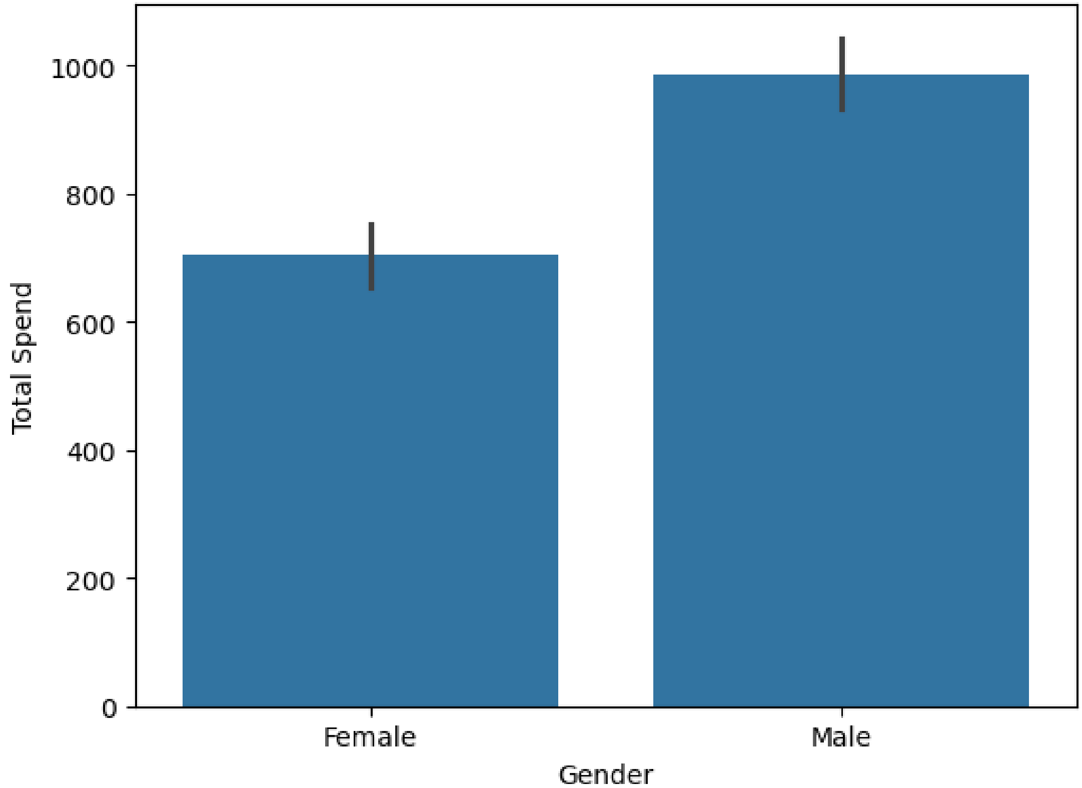

Fig 5. Average Total Spend by Gender

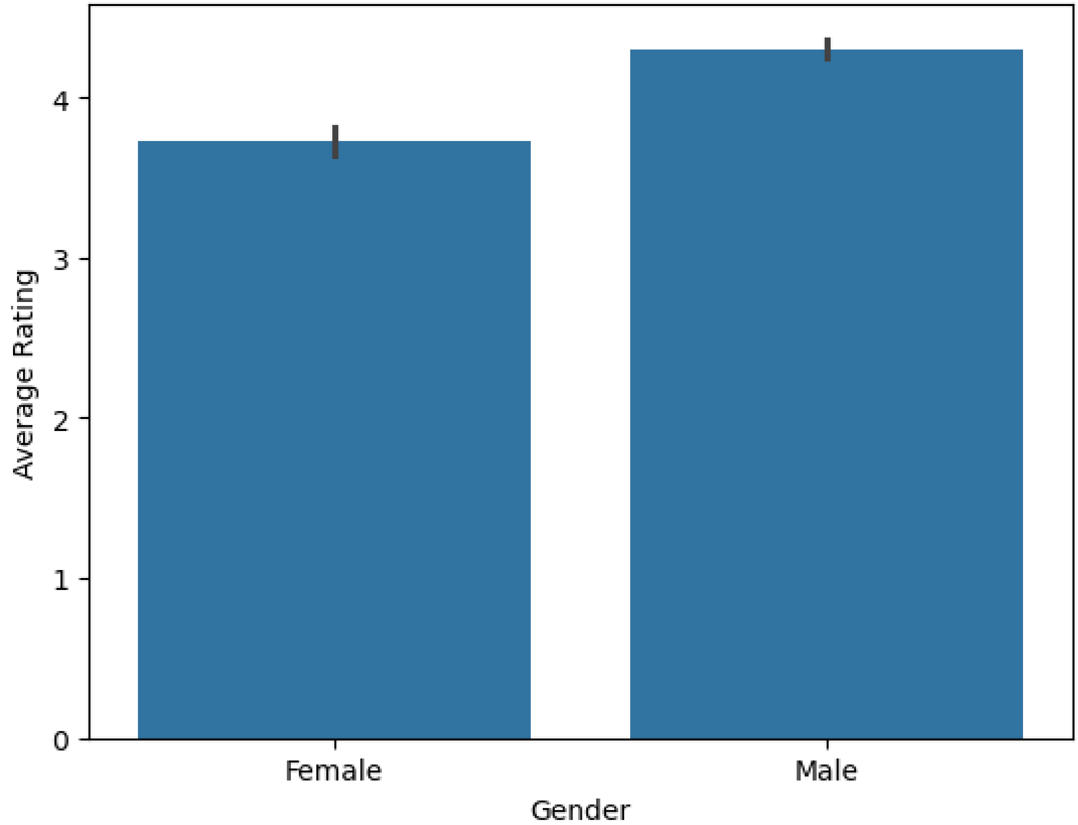

Fig 6. Average Rating by Gender

Metric	Male	Female	Difference
Avg. Total Spend	$986.93	$703.83	**+$283.10 (Male)**
Avg. Items Purchased	14.44	10.76	**+3.68 (Male)**
Avg. Rating	4.31	3.73	**+0.58 (Male)**

□ **Key Insight:** Male customers significantly outspend female customers (avg. $986.93 vs $703.83), purchase more items (14.44 vs 10.76), and leave higher satisfaction ratings (4.31 vs 3.73). This gap is worth digging into — it could mean the product catalogue or marketing language resonates more with male shoppers, or simply that male customers in this dataset have higher disposable income. Either way, there is a real opportunity to better understand and serve female customers, who make up roughly half the base but are currently underperforming on every key metric.

3.2 Membership Type Segmentation

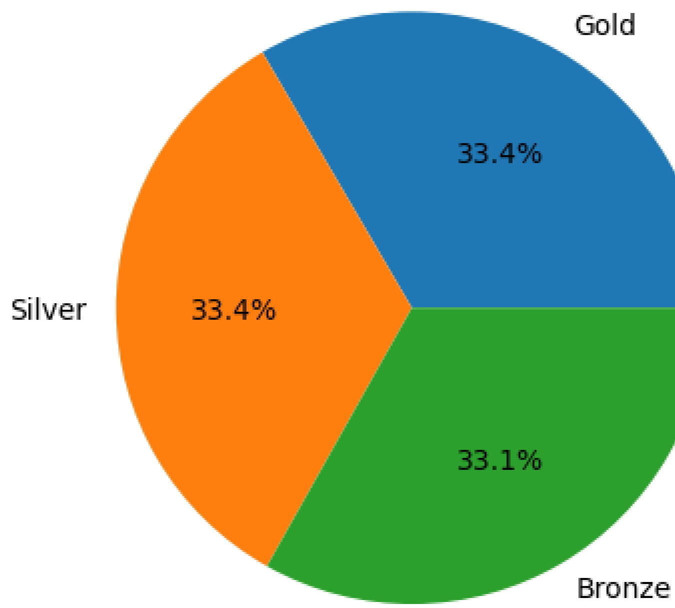

Fig 8. Customer Distribution by Membership

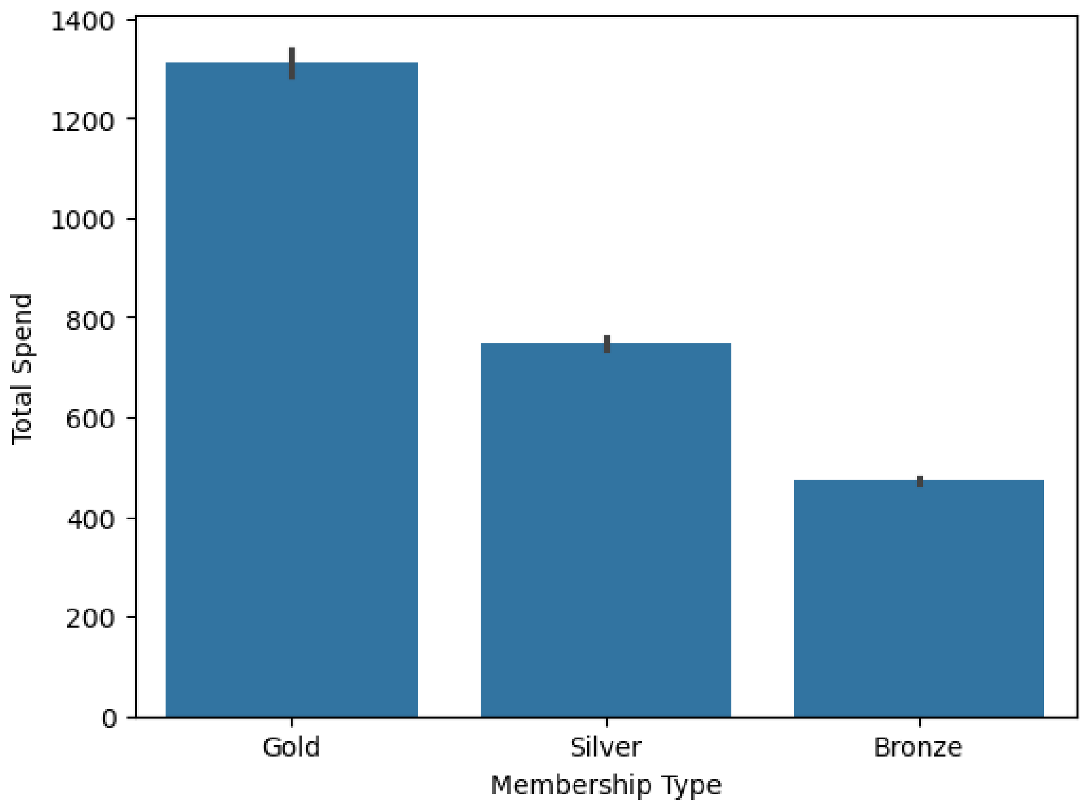

Fig 9. Avg. Total Spend by Membership Type

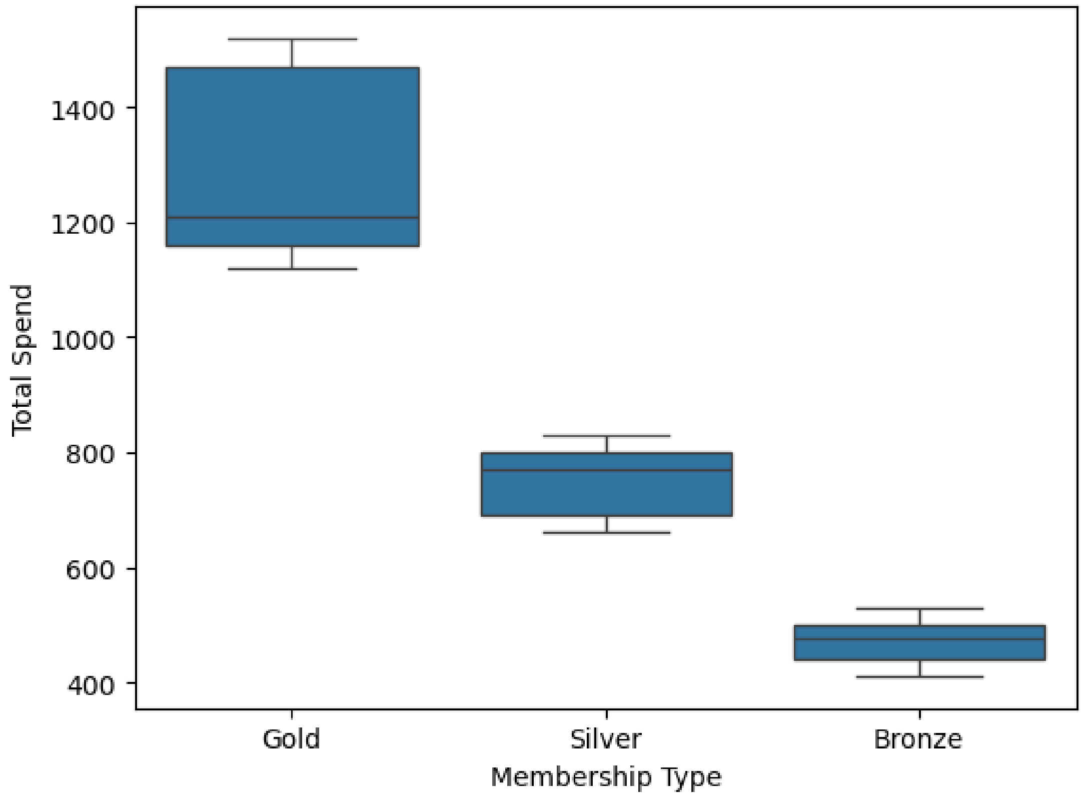

Fig 10. Total Spend Distribution by Membership Type

 Key Insight: Gold members are clearly the crown jewel of this business — they spend nearly double Silver members and triple Bronze members, and they are the most satisfied (4.68/5). The pattern is unmistakable: the higher the tier, the more engaged and valuable the customer. Bronze members, on the other hand, tell a different story — the lowest spend, the lowest satisfaction score (3.33), and the heaviest concentration in the Medium Risk churn bucket. This group needs attention now, not later. A targeted upgrade incentive (e.g., discounts to move from Bronze to Silver) could unlock meaningful revenue.

3.3 Age Distribution

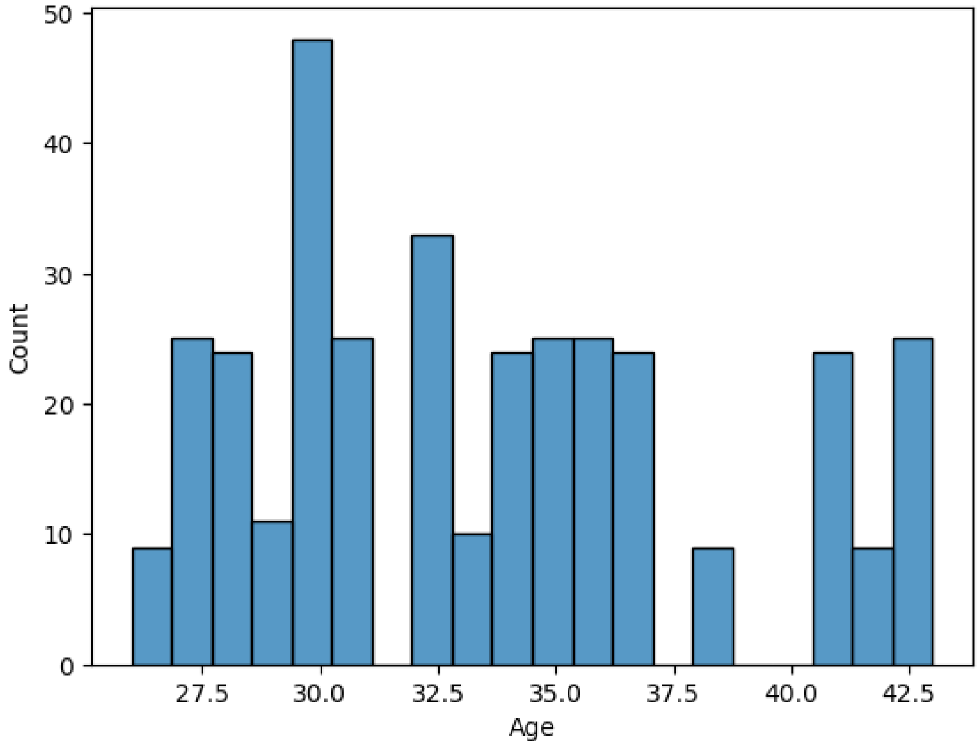

Fig 11. Age Distribution of Customers

Customers aged 28, 29 and 30 are the highest spenders and have the highest purchase frequency. This is not surprising — this age group tends to be in early-to-mid career stages with growing disposable income and a habit of online shopping. What makes this demographic especially valuable is that they are still building brand loyalty, meaning the business has a real window to lock them in through premium experiences, personalized recommendations, and loyalty rewards before competitors do.

3.4 Geographic Segmentation

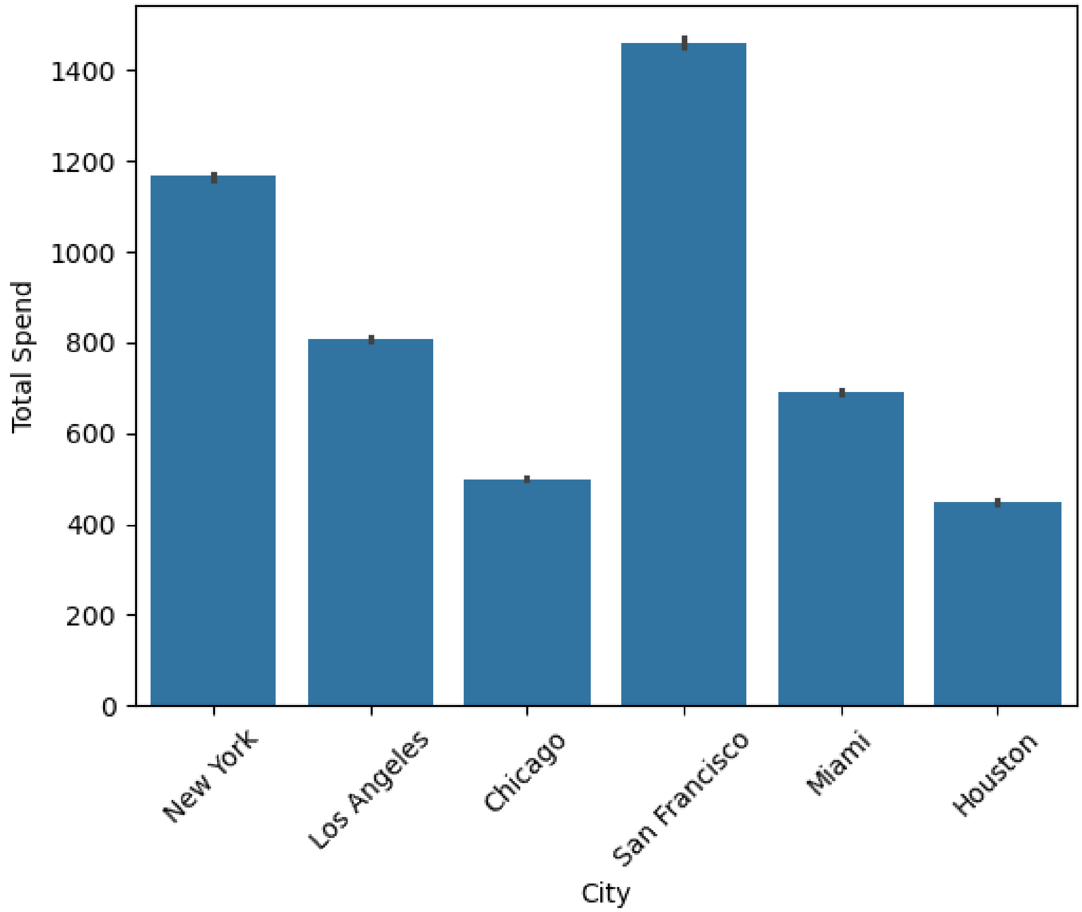

Fig 12. Average Total Spend by City

City	Avg. Spend	Relative Performance	Priority
New York	$1,165.04	⭐ Highest	Retain VIPs
San Francisco	$1,050+	⭐ High	Expand
Chicago	Below avg.	▯ Lagging	Re-engage
Houston	$446.89	▯ Lowest	Urgent Action

▯ **Key Insight:** The geographic picture is stark — New York customers spend $1,165.04 on average while Houston customers spend just $446.89, a gap of over 160%. This is not just a number; it represents a large slice of potential revenue that the business is leaving on the table. San Francisco performs strongly too, suggesting that high-income, tech-savvy urban markets are where this platform naturally thrives. The underperformance of Houston and Chicago likely reflects a combination of weaker marketing presence, product-category mismatch, and lower brand awareness. A city-specific campaign with local relevance — not a one-size-fits-all approach — is what these markets need.

3.5 Satisfaction Level Distribution

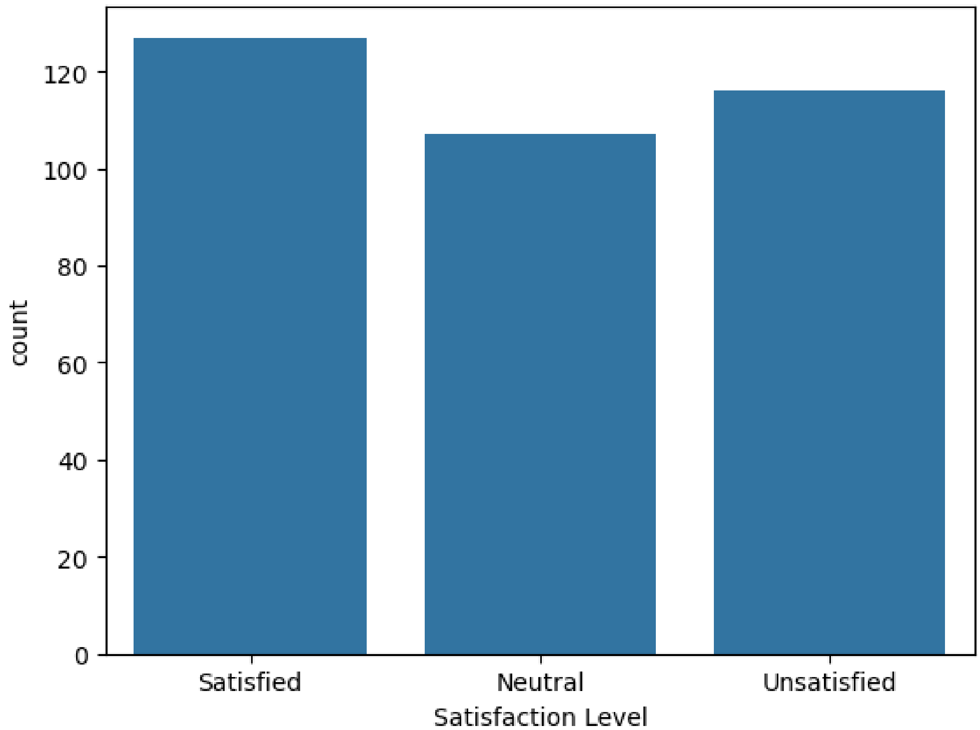

Fig 13. Customer Satisfaction Level Counts

The satisfaction distribution tells a reassuring story at first glance — most customers are Satisfied or Neutral. But the detail behind the Unsatisfied group is where the real signal lives. Unsatisfied customers cluster heavily among Bronze and Silver members in Chicago, Houston, and Miami. This is not a coincidence — these are the same people who spend less, engage less frequently, and are most at risk of churning. Satisfaction and spending reinforce each other: when customers feel good about their experience, they spend more; when they do not, they quietly drift away. Addressing dissatisfaction in these segments is not just a goodwill exercise — it is a direct revenue lever.

4. Retention Strategies & Churn Analysis

4.1 Churn Risk Classification

A new Churn Risk feature was developed based on Days Since Last Purchase to classify customers into actionable risk tiers. This is one of the most practical signals available — a customer who bought last week behaves very differently from one who bought two months ago, even if everything else looks the same on paper:

- Low Risk (Active): ≤ 30 days since last purchase
- Medium Risk (At-Risk): 31–90 days since last purchase
- High Risk (Churned): > 90 days since last purchase

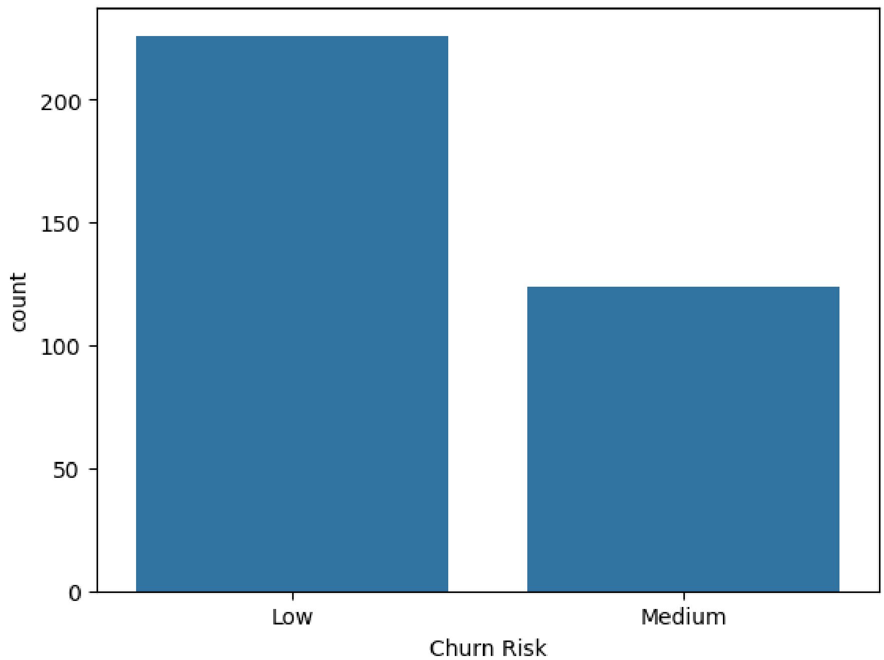

Fig 14. Churn Risk Distribution

226 Low Risk (Active)	✓ 64.6% of base	**124** Medium Risk (At-Risk)	⚠ 35.4% of base

4.2 At-Risk Customer Profile (Medium Risk)

Medium Risk customers show meaningfully lower spending and satisfaction compared to active (Low Risk) customers. Bronze members are heavily concentrated in this tier. Importantly, these customers have not yet fully disengaged — they are in a window where a well-timed, personalized outreach can genuinely bring them back. Waiting too long risks moving them into the High Risk bucket, which is far harder and more expensive to recover from.

Metric	Low Risk	Medium Risk

Avg. Total Spend	$969.30	$619.52
Avg. Satisfaction Rating	4.18 / 5	3.73 / 5
Primary Membership	Gold	Bronze / Silver
Items Purchased (avg.)	Higher	Lower

4.3 Personalized Retention Tactics

Based on the segmentation and churn analysis, the following targeted strategies are recommended. The key principle here is personalization — a generic discount email sent to everyone is far less effective than a message that feels like it was written for that specific customer based on their purchase history, tier, and location:

Segment	Risk Level	Strategy	Key Tactics
Bronze & Silver (Medium Risk)	□ Medium	Re-engagement Campaign	Personalized 'We miss you' email, limited-time discount, short feedback survey
Gold Members (Low Risk)	✅Low	VIP Loyalty Enhancement	Exclusive early access to sales, surprise gifts, dedicated support, referral bonuses
Houston & Chicago (Underperforming)	□ Medium–High	Geo-Targeted Marketing	Local influencer partnerships, city-specific promotions, free shipping days to stimulate purchases
All Medium Risk	□ Medium	Personalized Reactivation	Abandoned cart reminders, product recommendations based on purchase history, new arrival alerts

5. Conclusion & Recommendations

The analysis surfaces three clear action priorities for the business:

1. Protect and reward the Gold member base — particularly in New York and San Francisco. These customers generate the most revenue and are already highly satisfied. The risk is complacency: if they feel undervalued or discover a better alternative, they are the most expensive customers to replace. Double down on VIP perks, early access, and personal touches that make them feel genuinely special.

2. Re-engage the 35.4% of Medium Risk customers immediately — this is the biggest single lever for revenue recovery. These customers have not given up yet; they just need a reason to come back. A well-crafted, personalized re-engagement campaign that acknowledges their absence, offers real value, and asks for feedback can convert a significant portion of this group into active, loyal customers.

3	Invest seriously in underperforming markets like Houston and Chicago. The 160%+ spending gap versus New York is not a ceiling — it is an opportunity. These cities have customers on the platform already; they are just not being served in a way that resonates. Local influencer partnerships, city-specific promotions, and campaigns that reflect local culture and preferences can meaningfully close this gap over time.

The data has told a clear story. The business has strong foundations — engaged Gold members, active high-spending cities, and a satisfied core base. But it also has real vulnerabilities: a third of customers drifting toward churn, underperforming markets, and a Bronze tier that feels disconnected. By acting on these insights with targeted, human-centered strategies, the company can turn these vulnerabilities into its next phase of growth.